



Application-Centric Quantum Supercomputing

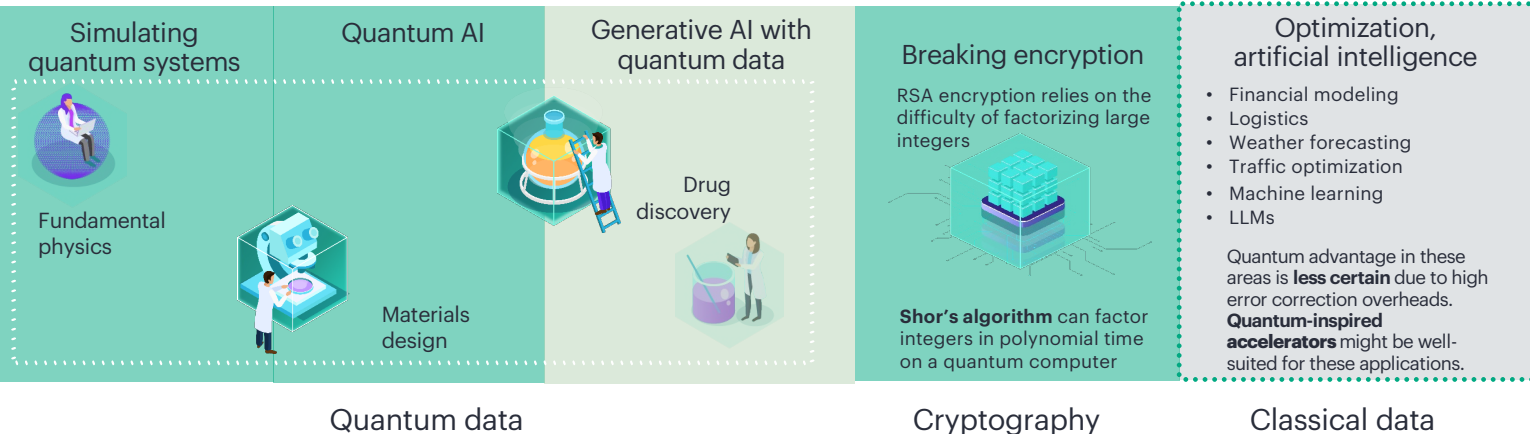
HPE Quantum

Presented by Namit Anand and Michael Ferguson

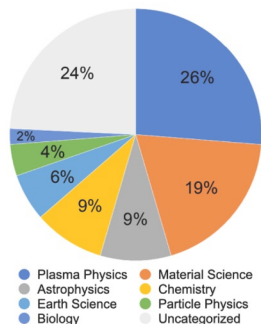
openQSE seminar, July 2026

What are quantum computers good for?

Our primary focus in this talk will be on the *quantum data* applications (first three columns).

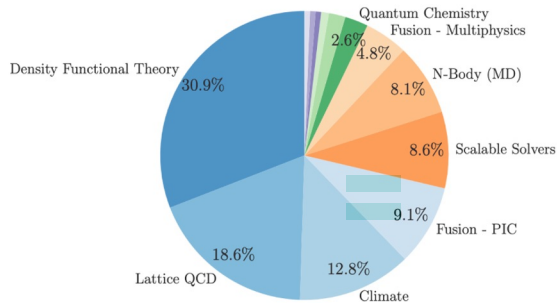


What can we learn from current HPC workloads?



500 million core-hours of computing workload on LANL's Chicoma in 2022

[arXiv:2406.06625 \(2024\)](https://arxiv.org/abs/2406.06625)



Workload distribution at NERSC HPCs (e.g., Perlmutter) across different categories

[arXiv:2509.09882 \(2025\)](https://arxiv.org/abs/2509.09882)

- 30-50% of the open-science simulations focus on **strongly-interacting electronic systems**.
- These problems are often classically intractable but can be simulated efficiently on quantum computers.

- Novel **quantum algorithms** will *fundamentally* change this workload.

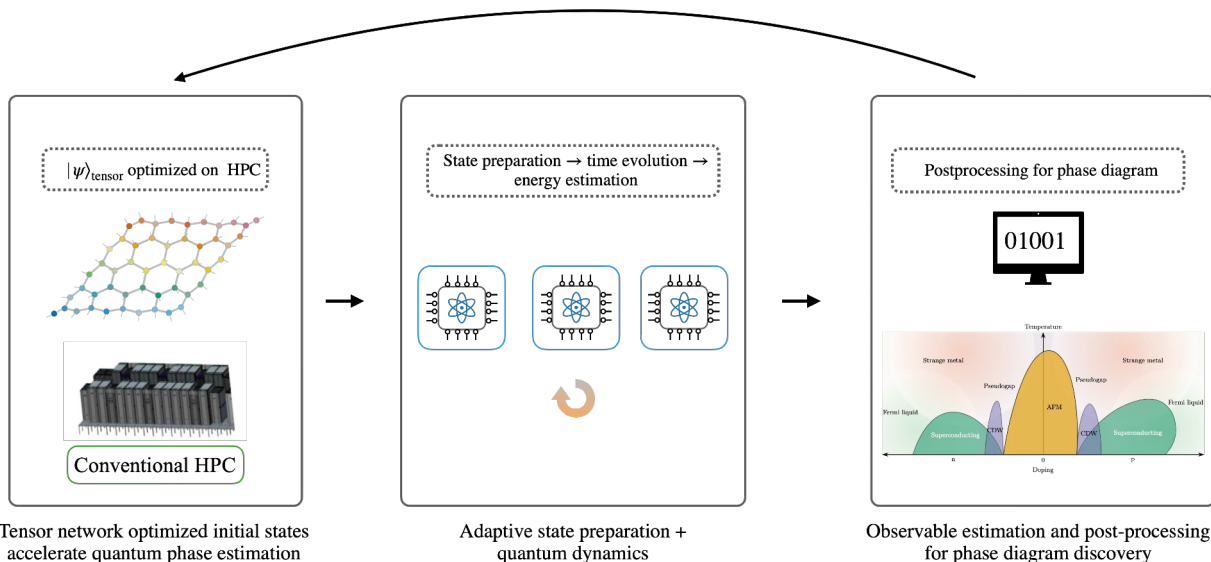
Accelerating quantum simulation



Accelerating typical quantum simulation workflows

(0) Classical Pre-processing (1) State preparation, (2) Quantum Dynamics, (3) Observable Estimation, (4) Classical Post-processing.

We are interested in algorithms and architectures that can accelerate all five steps.



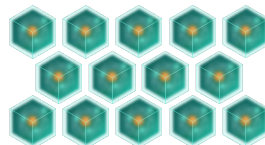
Fault-tolerant quantum computation (FTQC) vs Noisy intermediate-scale quantum (NISQ)

- Noise in current hardware limits the largest quantum computations that one can run to $\sim \log(n)$ -depth (where n is the number of qubits).
- Quantum Error Correction (QEC) reduces errors by using more qubits to encode information + encoding operations.
- New challenges: e.g., no universal fault-tolerant gates [Eastin–Knill theorem]
- Rotation gates need to be approximated with long (~ 300 gates) sequences of other gates, and T gates need to be provided separately (e.g. with magic state distillation and injection)
- Key distinction from NISQ: minimizing T gates as opposed to CNOTs.
- Real-time decoding requires significant classical compute. We see the need for additional classical compute.
- Compiling quantum algorithms will be different: approximate rotation gates, schedule magic state distillation, lattice surgeries, etc.

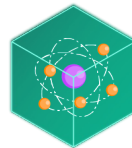
In summary, FTQC brings new challenges: scaling up qubits, different compilation stack, additional classical compute, etc.

Everything that follows in this talk will be FTQC (although many of our results apply to NISQ architectures as well)

Quantum Error Correction



Many noisy physical qubits



Logical qubit

Decoding QEC Codes (qLDPC) requires building flexible architectures

Tradeoffs between accuracy, latency, and hardware usage

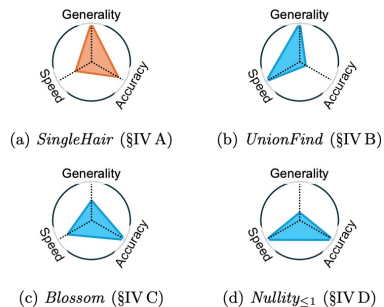


FIG. 7: Tradeoffs of different *Relaxer* finders. Generality, speed and accuracy cannot be achieved simultaneously given the NP-hardness of MWPF [5].

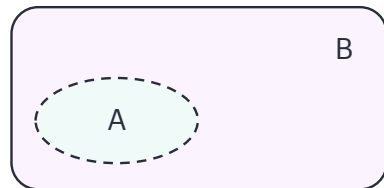
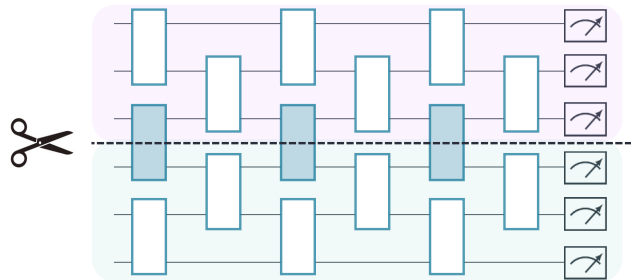
Wu et. al. "Minimum-Weight Parity Factor Decoder for Quantum Error Correction" [arXiv:2508.04969]

Panteleev & Kalachev 2021; Roffe et al. 2020 (BP+OSD); Hillmann et al., Nat. Commun. 2025 (LSD); Wolanski & Barber 2024 (Ambiguity Clustering, Riverlane); Müller et al. 2025 (Relay-BP, IBM); Maurer et al. 2025 (Relay-BP FPGA); Gong et al. 2024 (BP-GDG); Wang et al. 2025 (BP-SF); Sayginel et al. 2025 (AutDEC); Maan & Paler 2025 (Astra, npj Quantum Inf.); Chubb 2021 (tensor-network)

Distributed electronic structure simulation



What is the cost of distributing quantum simulation circuits?



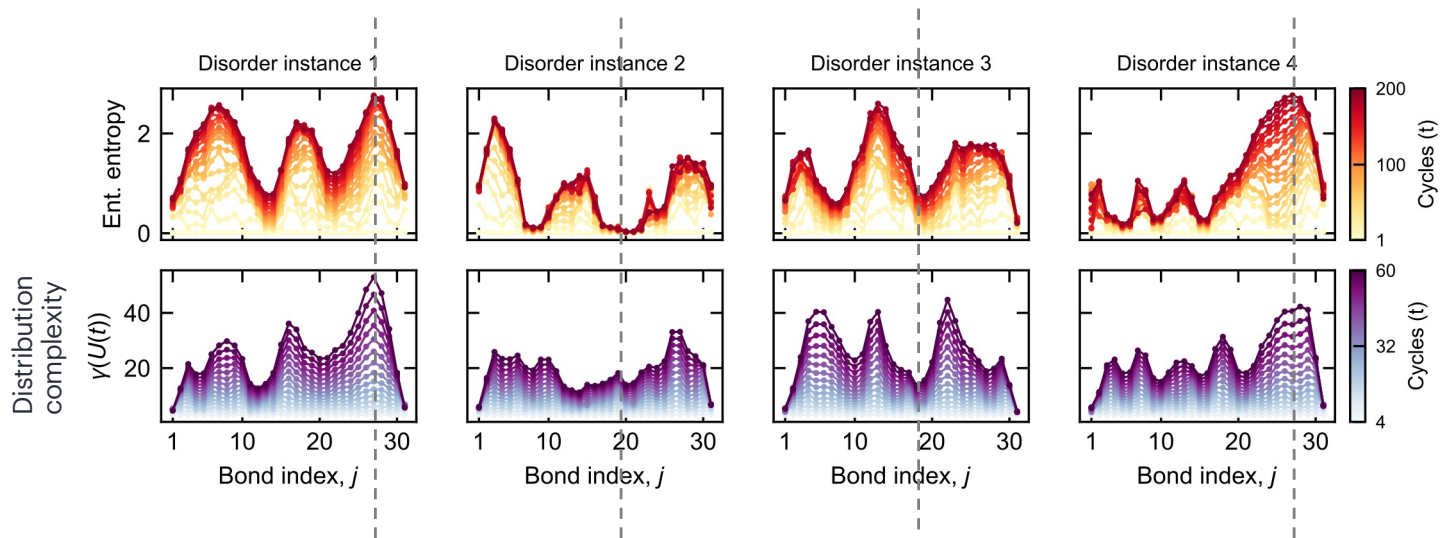
$$H = \sum_{i \in A} H_i + \sum_{j \in B} H_j + \sum_{k \in \partial A} H_k \quad \eta = \sum_{k \in \partial A} ||H_k||$$

- A clustered Hamiltonian can be simulated up to time step t with a sample overhead exponential in the norm of the boundary terms i.e., $O(\exp(4\eta t))$. Typically has exponential sample overhead [1]
- Finding optimal entanglement cuts allows us to reduce sampling overhead by orders of magnitude [2]

[1] Harrow, Lowe, Optimal Quantum Circuit Cuts with Application to Clustered Hamiltonian Simulation, PRX Quantum (2025)

[2] "Distributed Quantum Computing via Adaptive Circuit Knitting" [HPE Quantum, arXiv:2603.12411 (2026)]

Key insight: use entanglement to find optimal cuts



Spatial entanglement in 1D disordered spin-chain,
$$H = - \sum_{j=1}^{N-1} \sigma_j^x \sigma_{j+1}^x - g \sum_{j=1}^N \sigma_j^z - \sum_{j=1}^N h_j \sigma_j^x$$

Trotter Circuits for Electronic Structure Hamiltonians

- Consider a molecular Hamiltonian of the form, $\hat{H} = \sum_{pq} T_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} V_{pqrs} a_p^\dagger a_q^\dagger a_s a_r$ where the first term denotes electrons hopping on orbitals and the second term is electron-electron interaction.
- Quantum dynamics of these systems are modeled by quantum circuits with alternating layers of Gaussian unitaries (“orbital rotations”) and disordered diagonal unitaries (Coulomb interactions).
- What is the complexity of distributing quantum simulation over a network of QPUs?**

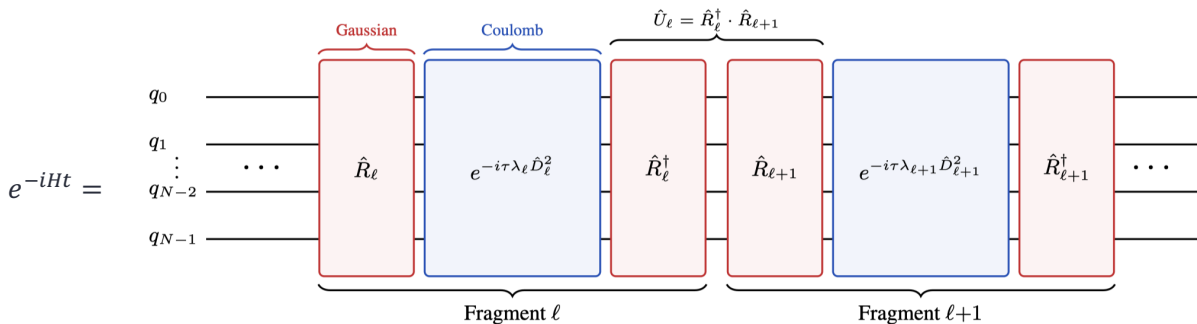
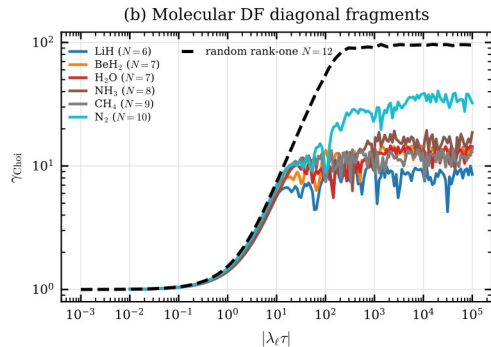
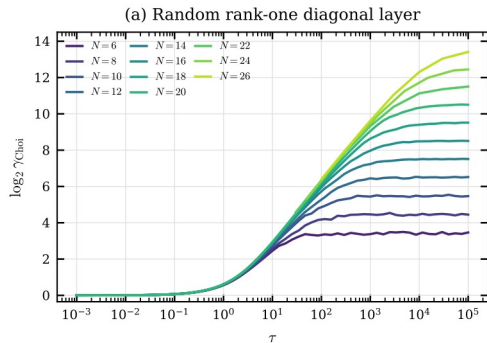
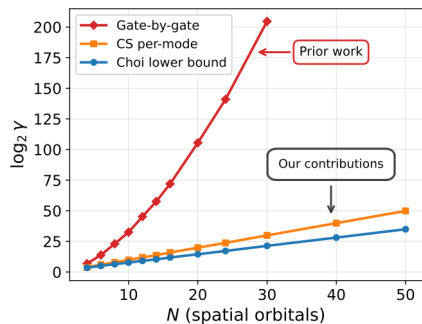


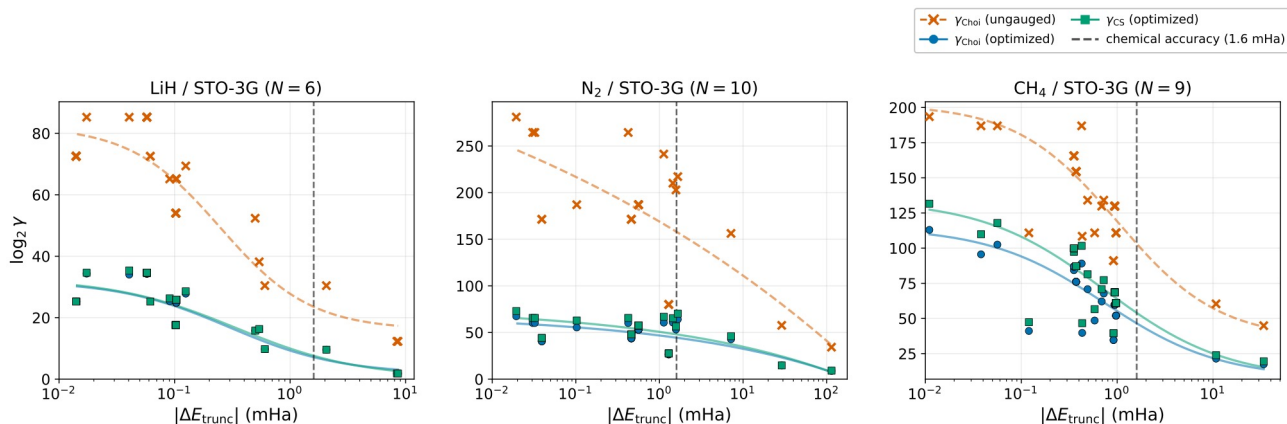
Figure 2: Trotter circuits for double-factorized electronic Hamiltonians. Quantum circuit for a single first-order Trotter step of the double-factorized Hamiltonian. Each fragment ℓ consists of a diagonal Coulomb interaction (blue), conjugated with a Gaussian orbital rotation (red). Adjacent fragment layers contain composite interfragment Gaussian rotations $\hat{U}_\ell = \hat{R}_\ell^\dagger \cdot \hat{R}_{\ell+1}$. The distribution complexity is therefore controlled by two types of objects: the compatibility of neighboring fragment bases and the diagonal interactions inside each fragment.

Comparison with prior work



- Our *entanglement-aware* compilation techniques show how to improve the classical interconnect cost from $\exp[N^2]$ to $\exp[N]$ in the worst-case and from N^2 to N for quantum interconnects
- For orbital rotations, we introduce an entanglement characterization method that computes its distribution cost with **exponential improvement** over previous methods (**2^N to $2N$ scaling**)
- For the diagonal Coulomb layers, we show that the distribution cost remains $O(1)$ for all Trotter time steps. This utilizes analytical techniques similar to that of *many-body localization* (MBL).

Example: correlation energy estimation on small molecules



Distribution complexity as a function of the truncation error for correlation energy estimation. We demonstrate the effect of removing weak fragments on distribution complexity for small molecules such as (a) LiH, (b) N₂, and (c) CH₄, all evaluated in STO-3G orbital bases. We show how to reduce the distribution cost by orders of magnitude while still retaining chemical accuracy.

Workflow for distributing quantum chemistry simulations

Input: Electronic structure Hamiltonian

$$H = \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{p,q=1}^N T_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{\alpha, \beta \in \{\uparrow, \downarrow\}} \sum_{p,q,r,s=1}^N V_{pqrs} a_{p\alpha}^\dagger a_{q\alpha} a_{r\beta}^\dagger a_{s\beta}$$

Transform to double-factorized form

$$H_{\text{DF}} = \frac{1}{2} \sum_{\ell=1}^L \lambda_\ell \hat{R}_\ell^\dagger \hat{D}_\ell^2 \hat{R}_\ell, \quad \hat{D}_\ell = \sum_m e_m^{(\ell)} a_m^\dagger a_m, \quad \hat{R}_\ell \in SO(N)$$

Obtain Trotter circuits (e.g., first-order)

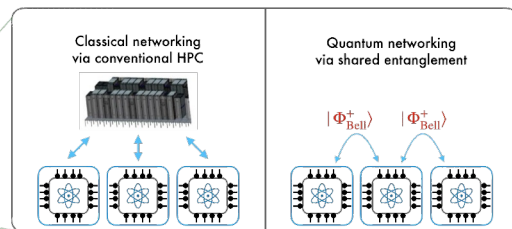
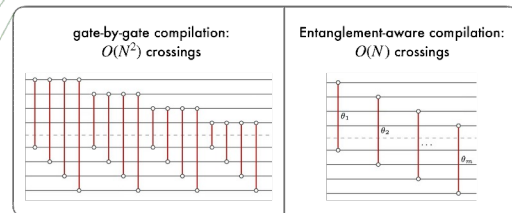
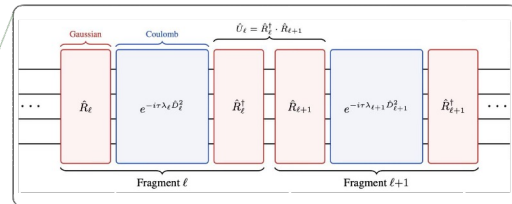
$$U_{\text{Trotter}}(\tau) = e^{-i\tau \hat{T}_{\text{eff}}} \prod_{\ell=1}^L \hat{R}_\ell^\dagger \exp\left(-\frac{i\tau \lambda_\ell}{2} \hat{D}_\ell^2\right) \hat{R}_\ell$$

Estimate distribution complexity

$$\gamma_{\text{Choi}}(\hat{R}; A|B) = 2 \prod_{k=1}^m \left(1 + \sqrt{1 - \sigma_k^2}\right)^2 - 1; \quad \gamma_{D, \epsilon|B} = \text{poly}(\lambda\tau)$$

Identify classical vs. quantum interconnects

Input: Total # of Trotter layers, available memory, fidelity of Bell-state generation, size of QPU, etc.



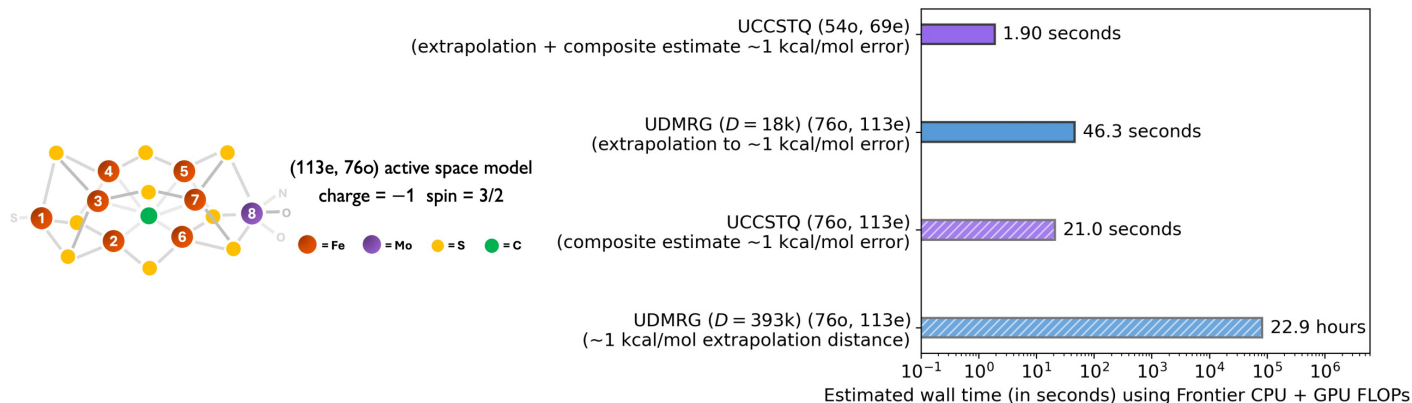
Electronic Structure: Implications for system design

- How long is the quantum compute?
E.g., FTQC on superconducting qubits with $1\mu\text{s}$ QEC clock cycle time, takes about 42 seconds.
 - Depends on the sub-problem size and accuracy requirements
- How long is the classical compute?
E.g., preparing “good” initial states for quantum phase estimation, takes about a few minutes.
- Interaction between the quantum and classical?
Classical ansatz to be sent to the QPU for state-preparation.
- How many quantum-classical iterations?
Depends on accuracy requirements
- Co-scheduled quantum and HPC is preferred since the algorithms/runtime is on the order of seconds to minutes.
- “Tight coupling”: Communications on the order of seconds is OK here

“Partially Fault-Tolerant
Quantum Computation
for Megaquop
Applications”
[arXiv:2603.13093]

Alg. N_{orb}			Time-optimal									Space-optimal					
			Logical Resources			Target Hardware			Desired Hardware			Target Hardware			Desired Hardware		
			# Log. qubits	# T gates		# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances
DF Qubitiz.	6	291	4.9×10^6	1.4×10^6	42.9 sec	23 25	1.0×10^6	42.9 sec	17 25	1.0×10^6	7.9 min	23 27	8.5×10^5	5.8 min	17 25		
	18	502	2.3×10^9	3.6×10^6	6.9 hrs	11, 29 31	2.5×10^6	6.9 hrs	23 31	2.6×10^6	2.9 days	11, 29 33	2.1×10^6	2.6 days	23 31		
	26	668	9.0×10^9	4.7×10^6	1.2 days	11, 31 33	3.5×10^6	1.2 days	23 33	3.6×10^6	19.9 days	13, 31 35	3.2×10^6	10.1 days	23 33		

FeMoCo: how much classical compute is needed for initial state preparation?



Figures from [1]

- The ground-states energies of strongly-correlated chemistry (e.g., Fe-S clusters) can be approximated by using extensive classical compute (DMRG + CC).
- Ref. [1] discussed 'hypothesized' costs for classical tensor network simulation for GSEE.
- **Classical optimization with approx. 1% to 10% of the compute above should suffice for a good initial state for QPE. This still needs to be prepared on the QPU!**

[1] "Classical computational simulation of the FeMo-cofactor model to chemical accuracy and its implications" [Huanchen Zhai, Chenghan Li, Xing Zhang, Zhendong Li, Seunghoon Lee, Garnet Kin-Lic Chan; arXiv:2601.04621 (2026)]

Fermi-Hubbard model

Simulating strongly-interacting electrons

- Many 'exotic materials' such as high-temperature superconductors, quantum spin liquids, topological matter has electrons strongly interacting with each other.
- Fermi-Hubbard models are a minimal way to understand the physics of these systems.
- Natural to simulate on quantum computers.
- Unfortunately, even the most optimal methods require $\sim 10^6$ T-gates and millions of qubits!
- Similar story for other applications such as NMR spectroscopy, catalyst design, quantum chemistry simulation

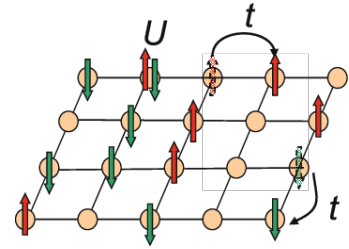


Figure from: [Hubbard model on Wikipedia](#)

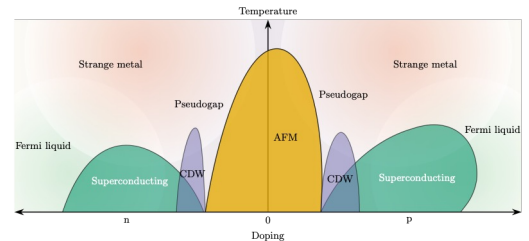
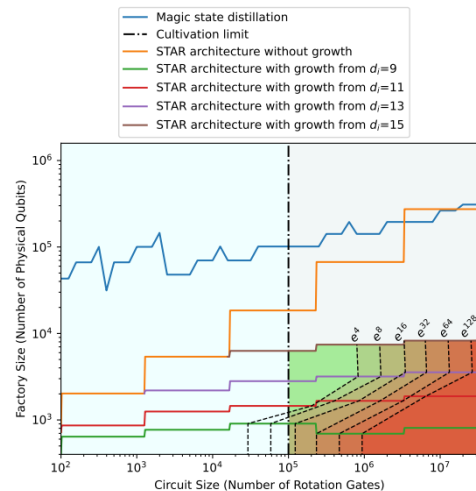


Figure from [7]

FTQC resources needed to simulate 2D Fermi-Hubbard

- Partial FTQC requires an order of magnitude fewer qubits ~ 100K instead of > 1M [1]
- Our work [1] focuses on simulating QPE via Trotterization for 2D Fermi-Hubbard with relative energy accuracy $\epsilon = 0.005L^2$
- We account for decoder delays, spacetime tradeoffs, scheduling, when comparing with full FTQC.

				Logical Resources								Target Hardware					
U/τ	L	ϵ		# Log. qubits	Max. Trotter	# Rot. gates	# Phys. qubits	# Phys. time	PEC free time	1 Trotter. clocks	Max. single-shot time	d_i	d_f				
4	4	0.08	65	150	3.4×10^4	4.7×10^4	21.3 min	12.1 min	314	561 ms	9	19					
	6	0.18	145	66	3.6×10^4	1.0×10^5	12.4 min	6.9 min	382	300 ms	9	19					
	8	0.32	257	37	3.8×10^4	2.3×10^5	8.6 min	5.0 min	442	205 ms	9	21					
	10	0.50	401	24	3.9×10^4	3.5×10^5	6.7 min	3.8 min	492	148 ms	9	21					
8	4	0.08	65	225	5.0×10^4	5.7×10^4	46.1 min	19.1 min	309	875 ms	9	21					
	6	0.18	145	100	5.5×10^4	1.3×10^5	24.3 min	10.5 min	378	476 ms	9	21					
	8	0.32	257	56	5.7×10^4	2.3×10^5	17.0 min	7.1 min	440	310 ms	9	21					
	10	0.50	401	36	5.9×10^4	3.5×10^5	12.8 min	5.4 min	490	222 ms	9	21					



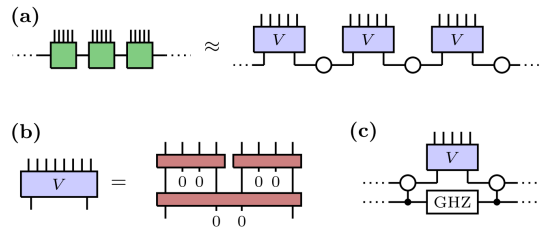
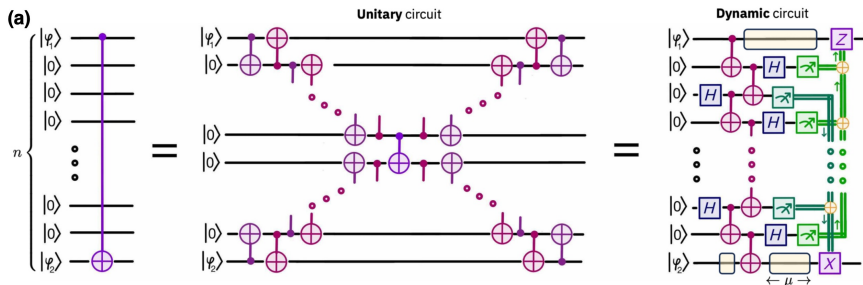
Hidden costs behind initial-state preparation?

- Quantum phase estimation (QPE) is the leading subroutine to learn ground-state/thermal state energy, which is then used to estimate other properties of the system.
- Most “quantum resource estimates” only discuss the cost of applying QPE circuits.
- However, this assumes that the initial state we are applying QPE on is already the true GS!
- In practice, for hard problems, we will only prepare an approx. GS, say with overlap $O(1/\text{poly}(n))$
- For QPE: this results in $O(\text{poly}(n))$ repetitions. We want to minimize this by optimizing the initial state “ansatz” on classical HPC first: this is the initial state problem.
- **Leading resource estimates show that preparing the initial state for strongly-correlated chemistry (e.g., [Fe-S] clusters) can require up to 10^{10} Toffoli gates!**

TABLE I. Extrapolated MPS overlaps as the output of the protocol described in Sec. III. The physical dimension of each subsystem for the MPS is taken to be $d = 4$, corresponding to $\{|\emptyset\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\}$. In the case of FeMoco, the overlap is with respect to a low-energy eigenstate, which may not be the ground state. The Toffoli and qubit counts are calculated using QUALTRAN [57].

System	Estimated $ \langle \text{MPS} \psi_0 \rangle $	Bond dimension	Spatial orbitals	MPS Toffolis	Qubits
Fe ₂ (III)Fe ₂ (II)	0.88	1000	36	42 200 000	359
Fe ₄ (III)	0.92	1000	36	42 200 000	359
FeMoco [50] MPS1	0.99	6000	76	1 360 000 000	833
FeMoco [50] MPS2	0.95	4000	76	733 000 000	682
FeMoco [50] MPS3	0.98	4000	76	733 000 000	682

Dynamic circuits for accelerating quantum state prep



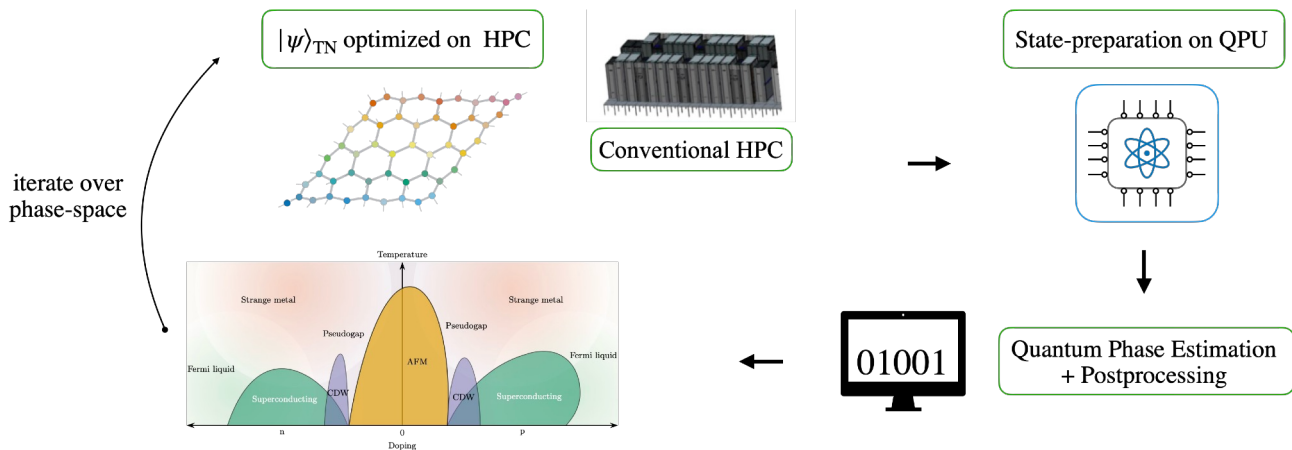
- Measurements and feedback can prepare arbitrary translation-invariant MPS (“tensor network states”) with an exponential speedup than purely unitary synthesis: from depth $O[\log(N/\epsilon)]$ to depth $O[\log \log(N/\epsilon)]$.
- Such MPS states provide an excellent approximation to the GS.
- Note: this measurement-and-feedback circuit is at the *logical* level.

[1] Efficient Long-Range Entanglement Using Dynamic Circuits [PRX Quantum 5, 030339 (2024)]

[2] Preparation of Matrix Product States with Log-Depth Quantum Circuits [Phys. Rev. Lett. 132, 040404 (2024)]

Workflow for discovering phase-diagrams with HPC-QCs

- Optimize tensor network ansatz on HPCs.
- Perform quantum phase estimation to learn GS energy
- Repeat to discover phase diagram, without state-preparation!
- Focus on quantum phase estimation cost as a proxy.



Fermi-Hubbard: Implications for system design

- How long is the quantum compute?
E.g., FTQC on superconducting qubits with $1\mu\text{s}$ QEC clock cycle time, takes about few seconds to minutes.
 - Depends on the problem size and accuracy requirements (e.g., relative accuracy vs absolute accuracy)
- How long is the classical compute?
E.g., preparing “good” initial states for quantum phase estimation, takes about a few minutes.
- Interaction between the quantum and classical?
Classical ansatz to be sent to the QPU for state-preparation.
- How many quantum-classical iterations?
Depends on accuracy requirements
- Co-scheduled quantum and HPC is preferred since the algorithms/runtime is on the order of seconds.
- “Tight coupling”: Communications on the order of seconds is OK here

"Partially Fault-Tolerant
Quantum Computation
for Megaquop
Applications"
[arXiv:2603.13093]

U/τ L ϵ				Time-Optimal									Space-Optimal								
				Logical Resources		Target Hardware			Desired Hardware			Target Hardware			Desired Hardware						
						# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances	# Phys. qubits	Phys. time	QEC code distances				
4	8	0.32	162	2.4×10^6	8.9×10^5	21.2 sec	21 25	7.0×10^5	21.2 sec	17 25	5.7×10^5	1.9 min	23 25	5.1×10^5	2.9 min	17 25					
	10	0.50	252	2.0×10^6	1.1×10^6	17.7 sec	21 25	9.4×10^5	17.7 sec	17 25	8.3×10^5	59.4 sec	21 25	7.4×10^5	2.4 min	17 25					
	16	1.28	642	1.7×10^6	2.1×10^6	15.0 sec	21 25	1.9×10^6	15.0 sec	17 25	1.9×10^6	33.1 sec	23 25	1.8×10^6	2.0 min	17 25					
	20	2.00	1002	1.6×10^6	3.1×10^6	14.4 sec	21 25	2.9×10^6	14.4 sec	17 25	3.1×10^6	14.4 sec	21 25	2.7×10^6	1.9 min	17 25					

Mid-Circuit Classical Computations

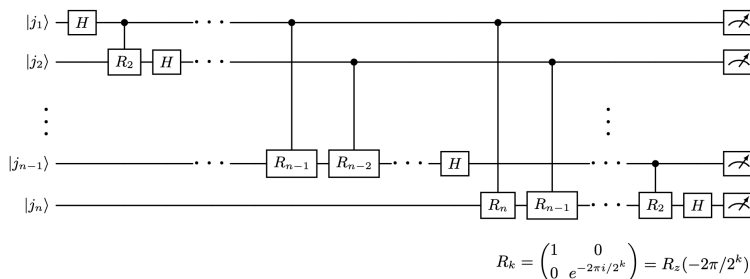
QFT
Quantum Oracle Sketch
QSVTF and Future Algorithms

Quantum Fourier Transform (QFT) Followed by Measurement

QFT is the **most important quantum** subroutine of all time. It enables almost all exponential speedups.

Many algorithms use an inverse QFT (QFT[†]) followed by measurement (QFT+M) [1]

Note: QFT and QFT[†] are closely related because the controlled phase gates are symmetric & so control and target qubits can be swapped [1]



O(n) gates?

We were excited to discover an adaptive circuit that reduces the gate count from $O(n^2)$ to $O(n)$.

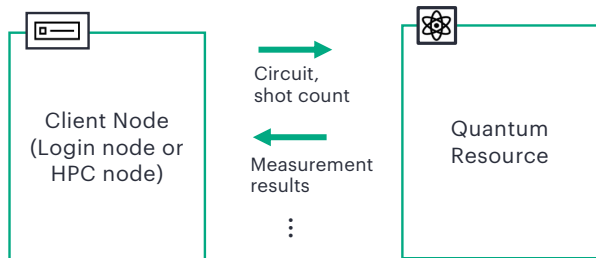
It turns out that we re-discovered a known result. However, it has implications for FTQC system design.

[1] Bäumer et al., [Quantum fourier transform using dynamic circuits](#), 2024

[2] Griffiths and Niu, [Semiclassical Fourier Transform for Quantum Computation](#), 1996

[3] Figure above from Ferguson et al., paper in preparation, 2026

An Initial FTQC System Design



Workflow program sends logical circuits to run, and the quantum resource runs them and returns results.

The circuits sent above can be adaptive.

That is, each circuit can include simple classical compute that runs **within the control electronics**.

Logical circuits will be compiled to physical circuits that include QEC & use dedicated hardware within the quantum resource for realtime decoding.

QFT+M for Qubitized Electronic Structure Simulation

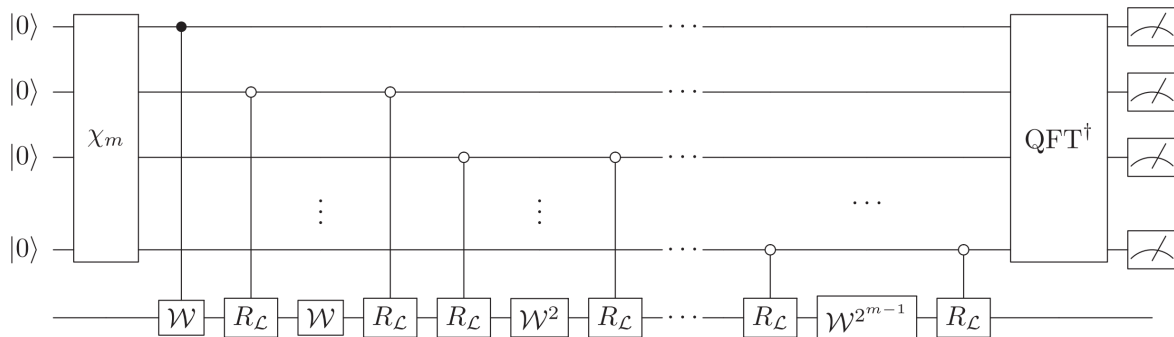


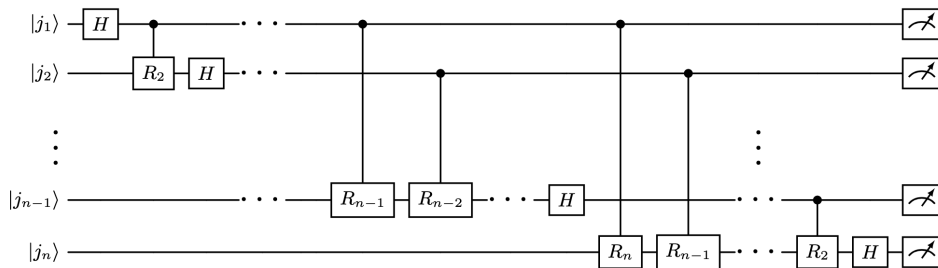
Figure 2 from [1]

Encoding Electronic Spectra in Quantum Circuits with Linear T Complexity has been a key paper in electronic structure simulation. The algorithm here ends with a $\text{QFT}^\dagger$ followed by measurement. **The resource estimation assumes it is done in $\mathcal{O}(n)$ gates, rather than $\mathcal{O}(n^2)$, using an adaptive circuit: the semi-classical procedure described in [2].**

This adaptive circuit would be impossible to run in the initial FTQC system design. Let's look at the semi-classical procedure to see why.

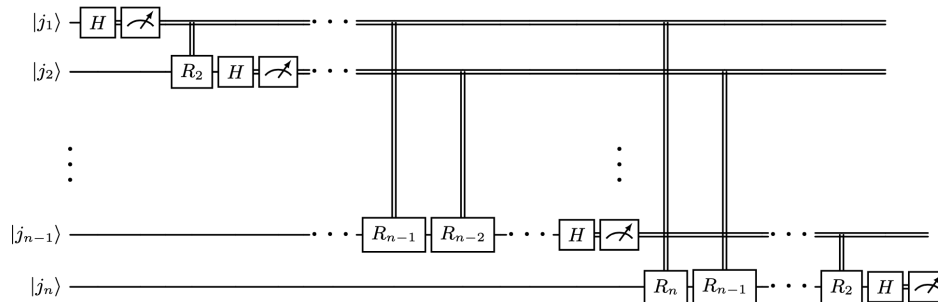
- [1] Babbush, et al., [Encoding electronic spectra in quantum circuits with linear t complexity](#), 2018
- [2] Griffiths and Niu, [Semiclassical Fourier Transform for Quantum Computation](#), 1996

Deriving the Semi-Classical Procedure (1/2)



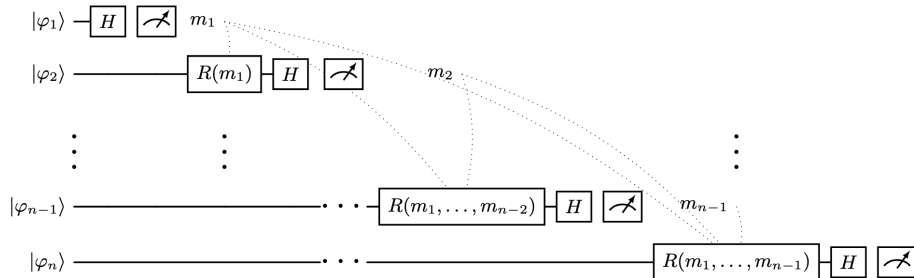
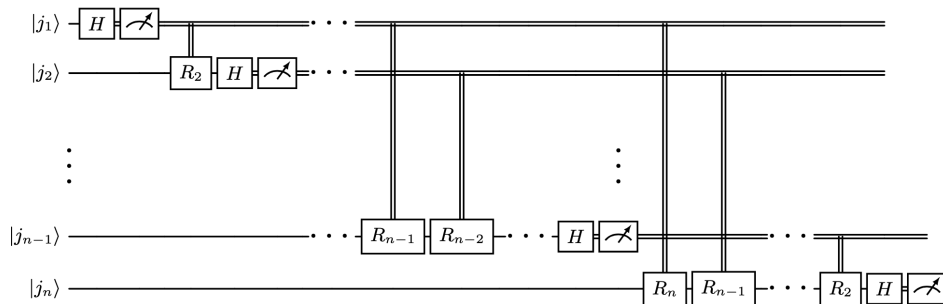
Here we have $\text{QFT}^\dagger$ followed by measurement. Note that there are $O(n^2)$ rotation gates.

Swaps are not shown & the bit order may be the reverse of standard.



Since measurement commutes with gates using the measured qubit as the control, we can move the measurement to the left to make all conditional gates classically controlled gates

Deriving the Semi-Classical Procedure (2/2)

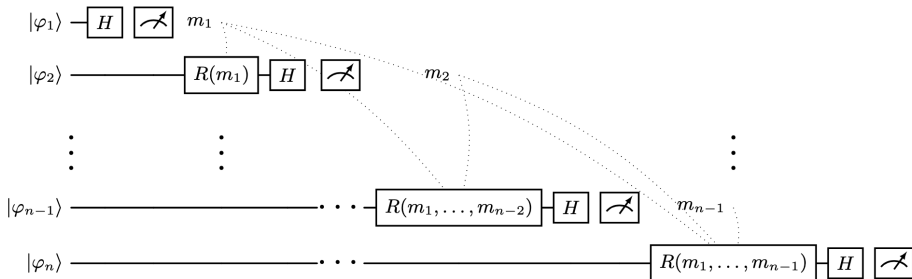


The rotation gates can be combined into rotation gates with an angle that depends upon all previous measurement results. This yields the same circuit that is described in [2] which has $n-1$ rotation gates.

[1] Figures above from Ferguson et al., paper in preparation, 2026

[2] Griffiths and Niu, [Semiclassical Fourier Transform for Quantum Computation](#), 1996

The Semi-Classical Procedure on FTQC

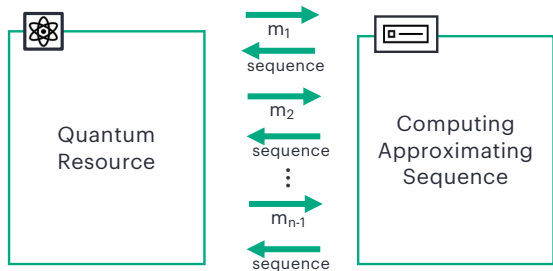


The angles for these rotation gates depend on measurement

FTQC systems approximate rotations with approximating gate sequences

Need to compute an approximating gate sequence mid-circuit!

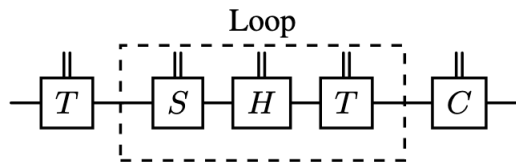
This cannot be done in control electronics. Therefore, it is not possible within the initial system design.



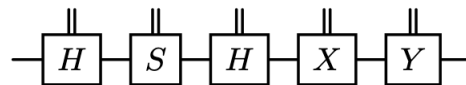
Externally-controlled Conditional Gates

We can adjust the system design to support this pattern:

- Control electronics support classically conditional gates where the condition comes from **outside the quantum resource** & mid-circuit measurements are sent externally as well
- We use the Matsumoto-Amano normal form for any single-qubit unitary: $(T|_{\epsilon})(HT|_{SHT})^*C$



Any single-qubit unitary can be represented as a loop with classical controls to indicate which unitary



Any single-qubit Clifford gate can be represented similarly

For example:

SHTSHTHTHTHTHTSHTSHTSHTSHX

Can be represented within 12 loops using 0 bits to skip unused operations:

0111111011011011011111011111110000001110

TSHTSHTSHTSHTSHTSHTSHTSHTSHTSHTSHTSHXY

Quantum Oracle Sketch

Proposed in recent work [1]; figure from [1] is reproduced to the right

Describes a method to learn a quantum state that approximates a large data set, among other things

Part of a class of quantum streaming algorithms: these are the only known quantum algorithms with **provable exponential space reduction**

What is the quantum circuit at the core of the algorithm, and what interaction does it need with classical resources?

As we shall see, it fits within the model of externally controlled conditional gates

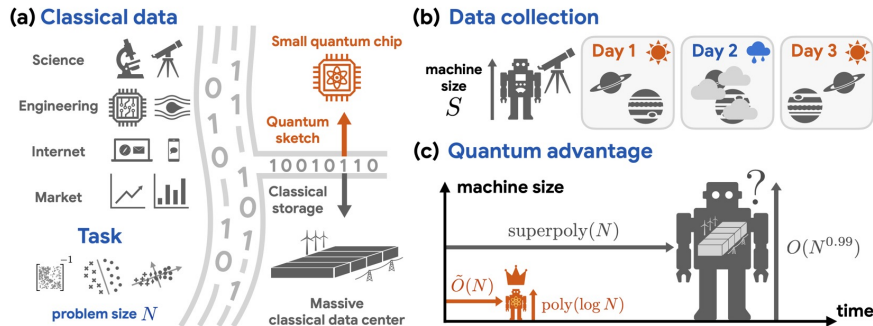


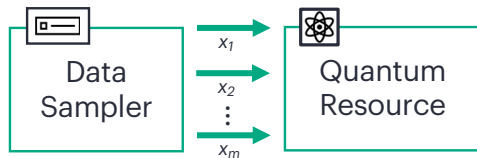
FIG. 1: **Overview of quantum advantage in processing massive classical data.** (a) We prove that a quantum computer can outperform exponentially larger classical machines in a wide range of classical data processing tasks, including solving linear systems, classification, and dimension reduction. (b) Our quantum algorithm enables coherent quantum queries to the noisy and evolving classical world. (c) For various classical data processing tasks with problem size N , a poly($\log N$)-size quantum machine can succeed in $\tilde{O}(N)$ time using quantum oracle sketching. In contrast, we prove that any classical machine, even with exponentially larger size $O(N^{0.99})$, cannot solve the same task unless given time super-polynomial in N . This exponential quantum advantage relies only on the principle of quantum superposition, independent of any computational complexity conjectures.

Sketching Quantum Oracle Sketch

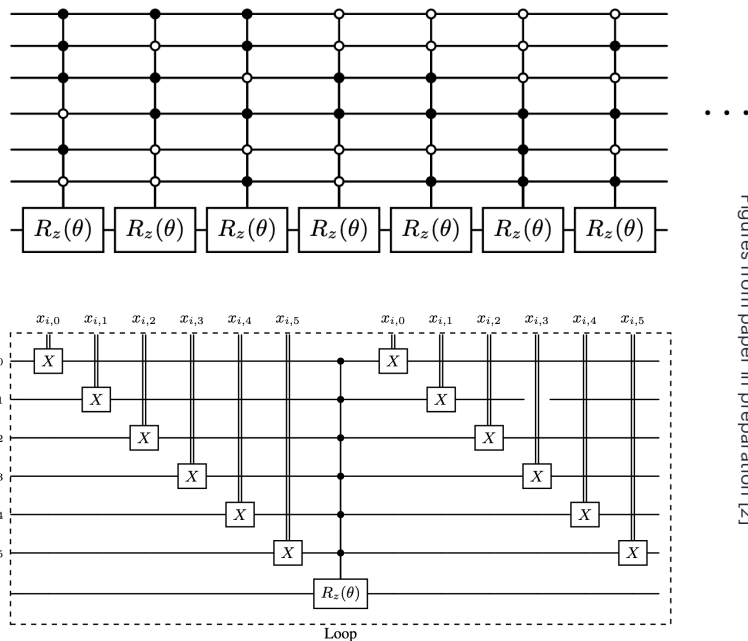
Algorithm 1 in [1] computes an approximation of a function f with $f(x_i) = y_i$; here x_i is an n -bit integer and y_i is 0 or 1

The circuit for Algorithm 1 applies a multi-controlled phase gate for each sampled data point x_i when $f(x_i) = 1$

- x_i forms the multi-control pattern
- θ is constant for each algorithm run



This circuit can be naturally represented with our proposal of externally controlled conditional gates; now the x_i values do not need to be known at circuit compilation time; note figure on the bottom loops m times



Figures from paper in preparation [2]

[1] Zhao et al., [Exponential quantum advantage in processing massive classical data](#), 2026

[2] Figures from Ferguson et al., paper in preparation, 2026

Feedforward QSVT and generalization

QSVT is a key building block within the most optimal quantum Hamiltonian simulation currently known

Authors of Feedforward QSVT propose a general framework for feedforward quantum algorithms, shown in the figure on the right from [1]

Definition 12. A feedforward quantum algorithm is a tuple $(\Theta, \mathcal{U}, \mathcal{M}, \mathcal{P}, \mathcal{C}, \theta_0)$. From an initial circuit structure parameter θ_0 , the circuit structure takes values in the parameter space Θ based on the record of measurement outcomes and previous circuit structures. The design policy $\mathcal{P}$ is derived by minimizing the cost metric $\mathcal{C}$.

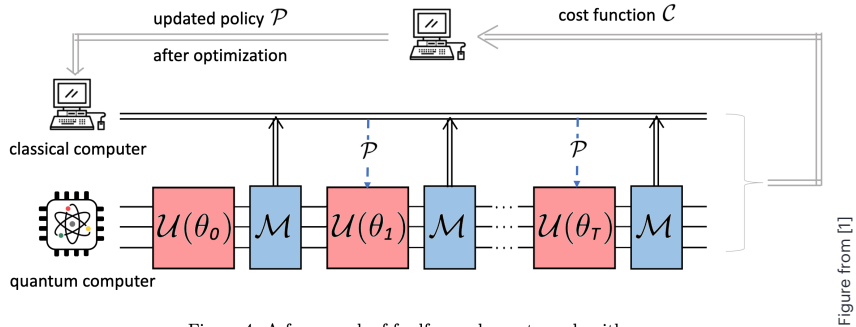


Figure 4. A framework of feedforward quantum algorithms.

Feed-forward quantum machine learning [2,3] is one algorithm that fits into such a scheme; it would run a classical optimizer to choose parameters & need to synthesize rotations

- [1] Dong et al., [Feedforward quantum singular value transformation](#), 2024
- [2] Carolan et al., [Variational Quantum Unsampling on a Quantum Photonic Processor](#)
- [3] Broughton et al., [TensorFlow Quantum: A Software Framework for Quantum Machine Learning](#)

Classical compute needs of application examples

	Estimated Quantum Circuit Runtime	Estimated Classical Runtime	Number of Quantum-Classical Iterations
Fermi-Hubbard (plaquette trotterization)	seconds	few seconds to minutes on 100 nodes	10 to 10,000
Fermi-Hubbard (qubitization)	hours	few seconds to minutes on 100 nodes	10 to 10,000
Distributed electronic structure	hours to days	few seconds to days on 1,000 nodes	10 to 100

Significant classical compute resources are required in some application phases
Compute requirement is often configurable & some applications only need modest HPC
HPC-QC communication on the order of seconds is acceptable for these applications

Logical Mid-Circuit Classical Compute Needs

Fermi-Hubbard: preparing an initial state with optimal depth

- Optimal state-preparation circuits use mid-circuit-measurement and feedforward
- Classical compute likely to be **simple enough for the control electronics**

QFT+M: compute 1 rotation sequence at a time

- Computing rotation sequence is **too much for control electronics**, but plausible with nearby programmable compute
 - E.g., we observed ~4 ms on a laptop, which may be fast enough for systems with a slower FTQC clock
 - It's very likely that a significantly faster version can be implemented to use parallel resources (multi-core CPUs, GPUs, or even multi-node)

Oracle sketching: apply a multi-controlled phase gate that depends on sampling a large data set

- **The data to be sampled will not fit into control electronics**
- Usefulness of oracle sketching depends on fast interaction and a fast FTQC clock

Future adaptive circuits: ?

- Cannot predict mid-circuit requirements of yet-to-be-invented algorithms; but **unlikely that all will fit into control electronics**
- We anticipate feed-forward learning circuits: mid-circuit running a classical optimizer and computing rotation sequences



Mid-Circuit Classical Compute

Latencies & Design Implications

QPU Modality	Approximate Gate Time	Approximate Coherence Times	Expected FTQC Clock
Superconducting	10-100 ns [1]	100 μ s – 1 ms [1]	1 μ s [2,3] – 10 μ s
Neutral Atoms	1 μ s – 500 μ s [4]	seconds [1]	900 μ s [4]
Trapped Ions	200 μ s [5] - 100 ms [1]	seconds to minutes [1]	6 ms [5]

	HPC Network	Commodity Network	Internet	Web page load	Carrier Pigeon	Scheduler Queue
End-to-End Latency	2-4 μ s	100 μ s	10 – 100 ms	seconds	hours - days	hours - days

Logical mid-circuit computations are feasible on HPC nodes if QC-HPC uses an HPC network
Any inefficiencies will slow application & contribute to cumulative logical error rate

[1] Buchs et al., [The Role of Quantum Computing in Advancing Scientific High-Performance Computing: A perspective from the ADAC Institute](#), 2025

[2] Google Quantum AI and Collaborators, [Quantum error correction below the surface code threshold](#), 2025

[3] [Alice & Bob's Quantum Computing Roadmap](#), 2024

[4] Zhou et al., [Resource Analysis of Low-Overhead Transversal Architectures for Reconfigurable Atom Arrays](#), ISCA'25

[5] Tripiier et al., [Fault-Tolerant Quantum Computing with Trapped Ions: The Walking Cat Architecture](#), 2026

Design for Flexibility

Q How much classical compute is needed and how tightly coupled is it?

A It depends on the application

We want a single quantum supercomputer design to support all of these applications

So, design for flexibility to support the wide variety of applications and continuing research. Goals:

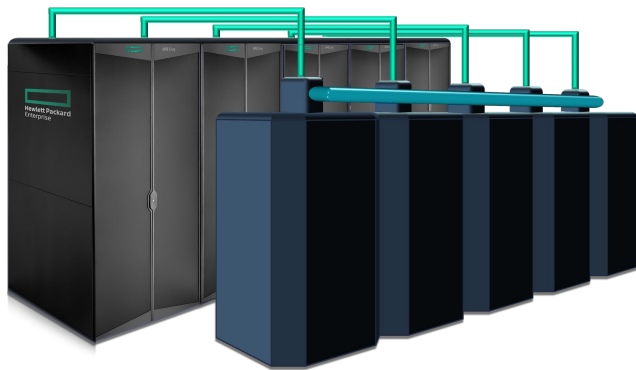
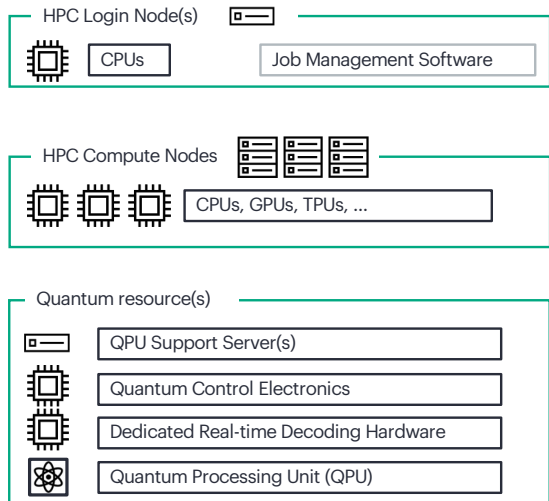
- Researchers should be able to explore novel adaptive quantum circuits with non-trivial classical compute requirements
- Researchers should be able to customize how their hybrid quantum-classical applications use available quantum and classical hardware
- Researchers should be able to work with their HPC center to explore using specialized classical hardware in hybrid quantum-classical applications
- Researchers should be able to try layered or alternative QEC strategies on real qubits

HPE's direction to meet these goals:

- Hardware: Quantum resources + co-located HPC nodes + a low latency network connection
- Software: Build upon OpenQSE, MQSS and **extend for mid-circuit classical compute on HPC nodes**



Systems Integration for Quantum Supercomputers



HPE is focused on quantum system integration